DATA MINING COURSE PROJECT

Detecting Fake News During the 2016-2018 U.S. Election Cycle

A Text Mining Analysis Using Topic Modeling, Sentiment Analysis, Word Frequency Visualization, and Predictive Classification

Tool: Orange 3.39 | Dataset: True and Fake News Sample (44,867 articles)

Winter 2026 | Data Analytics in Business Program

Table of Contents

1. Business Reason: The Problem I Set Out to Solve 3
   1.1 Why Orange 3.39? 3
   1.2 Future-Facing Application 3
2. Tool Documentation: How Orange 3.39 Was Used 4
   2.1 Backbone: Data Loading and Preprocessing 5
   2.2 Branch A: Word Cloud Analysis 6
   2.3 Branch B: Sentiment Analysis 6
   2.4 Branch C: Topic Modeling 7
   2.5 Branch D: Supervised Classification 8
3. Algorithm Selection: Why These Classifiers? 9
4. Data: Source, Structure, and Preparation 10
   4.1 Dataset Selection 10
   4.2 Dataset Structure 10
   4.3 Data Preparation 10
5. Results: What the Data Revealed 11
   5.1 Word Cloud Analysis: Vocabulary Fingerprints 11
   5.2 Topic Modeling: Thematic Structure of the Corpus 13
   5.3 Sentiment Analysis: Emotional Tone by Class 14
   5.4 Predictive Classification: Algorithm Performance 15
6. Lessons Learned: The Three W's 16
   6.1 What Went Well 16
   6.2 What Did NOT Go Well 17
   6.3 What Would Be Done Differently Next Time 17

1. Business Reason: The Problem I Set Out to Solve

The 2016 United States presidential election marked a turning point in the global conversation about misinformation. For the first time, the term 'fake news' entered mainstream political and media discourse as a documented, measurable phenomenon that shaped voter perceptions, drove social media engagement, and influenced electoral outcomes. Studies conducted after the election found that fabricated stories - many originating from politically motivated websites or foreign-operated influence operations - received millions of shares, often outperforming credible journalism in reach and engagement. The challenge, however, was not merely the existence of fake news. It was the speed at which it spread, the difficulty of distinguishing it from real news at a glance, and the lack of scalable tools to intercept or flag it in real time.

This project was motivated by that challenge. The business question driving my analysis is: can a machine learning system, trained on labeled examples of true and fake news from the 2016-2018 election cycle, learn to distinguish the two with meaningful accuracy - and in doing so, identify the linguistic, emotional, and topical fingerprints that separate misinformation from credible reporting? The answer has direct applications for social media platforms, news aggregators, fact-checking organizations, and election integrity agencies that need to evaluate the credibility of political content at scale and in near-real time.

1.1 Why Orange 3.39?

Orange 3.39 was selected as the analytical environment for this project for three primary reasons. First, it is an open-source, visual workflow platform that does not require programming expertise, making it appropriate for this project at this level. Its node-based canvas allows each analytical step to be inspected, documented, and reproduced without code. Second, its Text Mining add-on provides a fully integrated natural language processing pipeline - from corpus loading and preprocessing through topic modeling, sentiment analysis, and classification - within a single, consistent interface. Third, Orange's seamless integration between its text processing and machine learning widgets allows TF-IDF document vectors produced from the text pipeline to flow directly into classifier nodes, enabling a clean end-to-end workflow without data format conversion.

Alternative tools considered included KNIME and RapidMiner. KNIME proved to be challenging in the tuning of node settings. RapidMiner provides similar visual workflow capabilities but lacks the depth of text mining widgets available in Orange's Text Mining add-on, and also it had a limitation of 10,000 rows.

1.2 Future-Facing Application

The methods developed in this project are not just retrospective. The election cycle that produced this dataset - 2016 to 2018 - established many of the patterns that still characterize political misinformation today. A classification model trained on this data serves as proof of concept for several real-world deployment scenarios:

- **Social media moderation:** Flagging or deprioritizing articles that match the linguistic profile of fake news before they reach wide distribution.

- **News aggregator trust scoring:** Assigning credibility scores to articles in news feeds based on language patterns and topic composition.
- **Election integrity portals:** Providing voters and journalists with automated preliminary assessments of viral political claims during election seasons.
- **Fact-checking workflow support:** Prioritizing articles for human fact-checker review based on predicted fake-news probability scores.

2. Tool Documentation: How Orange 3.39 Was Used

The complete analytical workflow was built in Orange 3.39 with the Text Mining add-on installed. The workflow consists of a shared preprocessing backbone and four analytical branches. The figure below shows the complete workflow canvas as implemented.

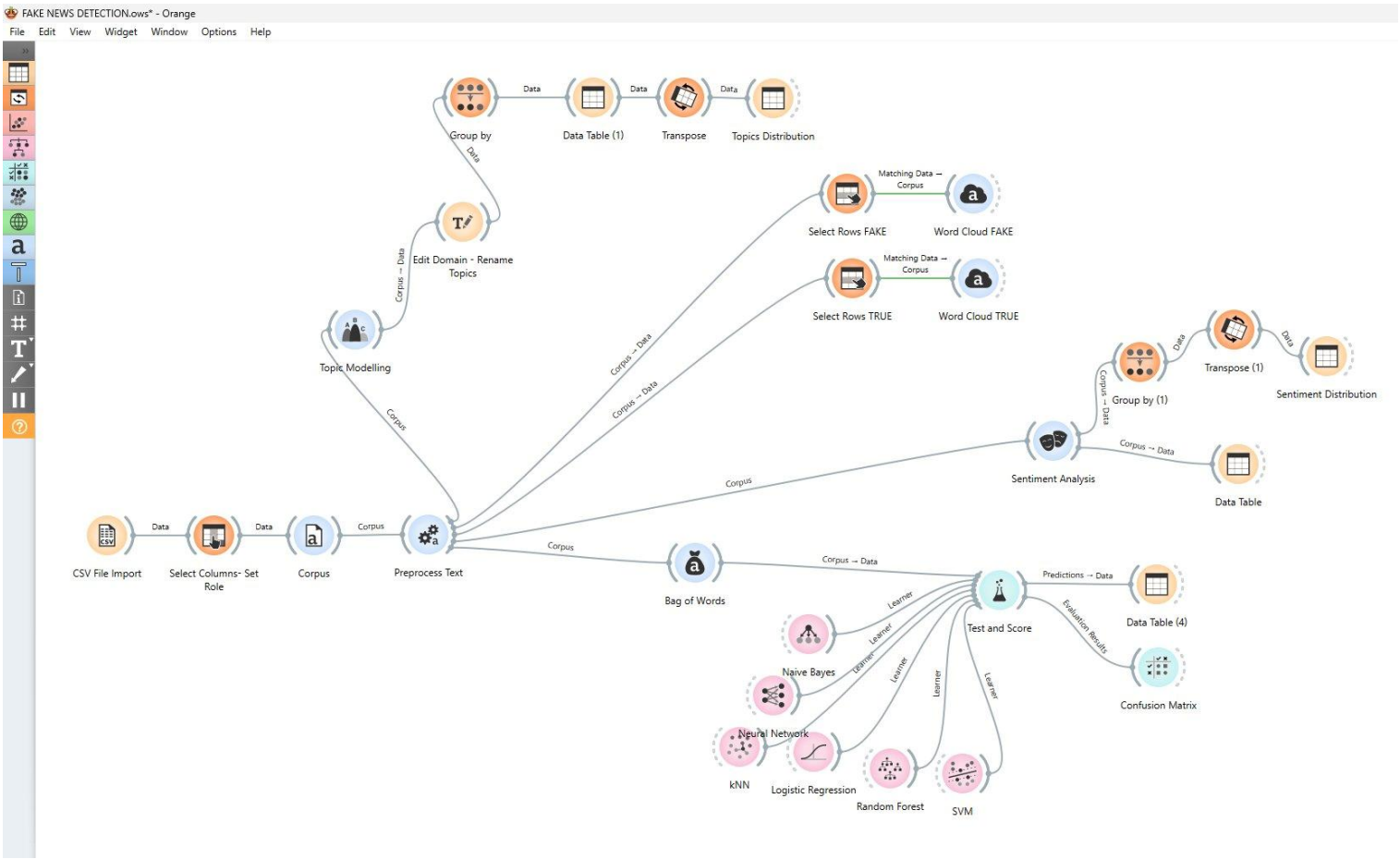
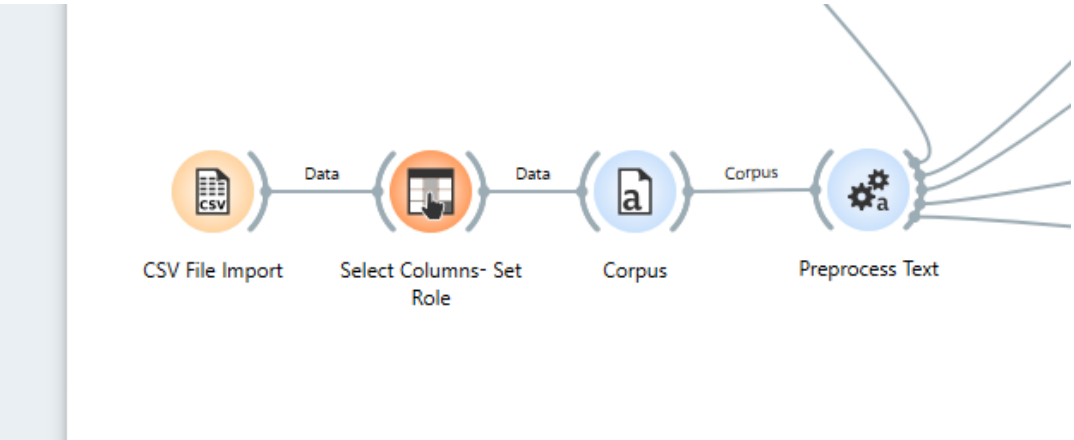


Figure 1. Complete Orange 3.39 workflow canvas for the Fake News Detection project.

2.1 Backbone: Data Loading and Preprocessing

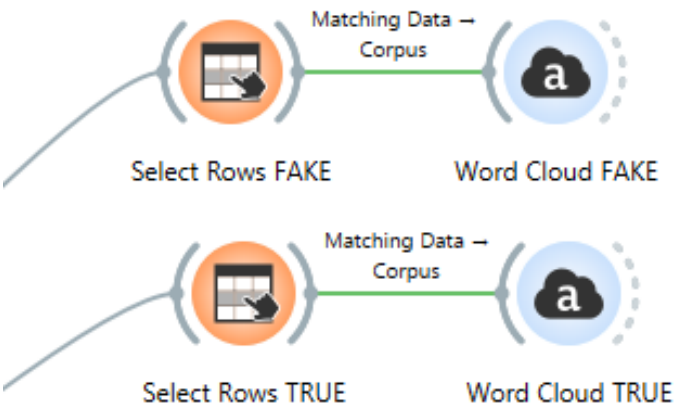
The workflow begins with three nodes that form the shared backbone used by all four branches.

Node	Configuration and Purpose
CSV File Import	Loaded fake_news_combined.csv (44,867 rows, 5 columns: title, text, subject, date, label). The 'label' column was set as the Target (class) variable.
Select Columns - Set Role	Used to confirm column roles and exclude non-text metadata from the corpus text field. Ensures only the 'label' column is the target.
Corpus	Converted the data table to an Orange Corpus object. 'text' was set as the used text feature; 'title' as the document identifier; 'label' as the class variable. Language set to English.
Preprocess Text	Applied: Lowercase transformation, URL removal, Regexp tokenization ([a-zA-Z]+), WordNet Lemmatization, English stopword filtering, document frequency filter (relative: 0.001-0.80), minimum token length of 3.



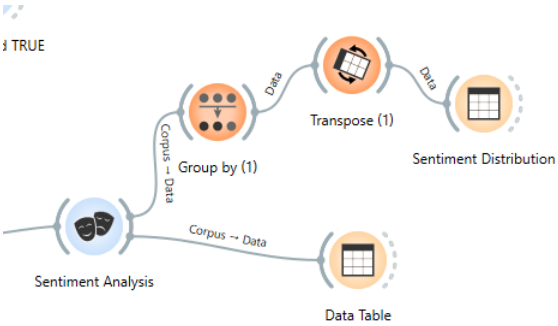
2.2 Branch A: Word Cloud Analysis

Node	Purpose and Configuration
Select Rows (FAKE)	Filtered corpus to label = 'Fake'. Connected from Preprocess Text.
Word Cloud (FAKE)	Displayed top 50 tokens from fake news articles. Larger text = higher frequency.
Select Rows (TRUE)	Filtered corpus to label = 'True'. Connected from Preprocess Text.
Word Cloud (TRUE)	Displayed top 50 tokens from true news articles for side-by-side vocabulary comparison.



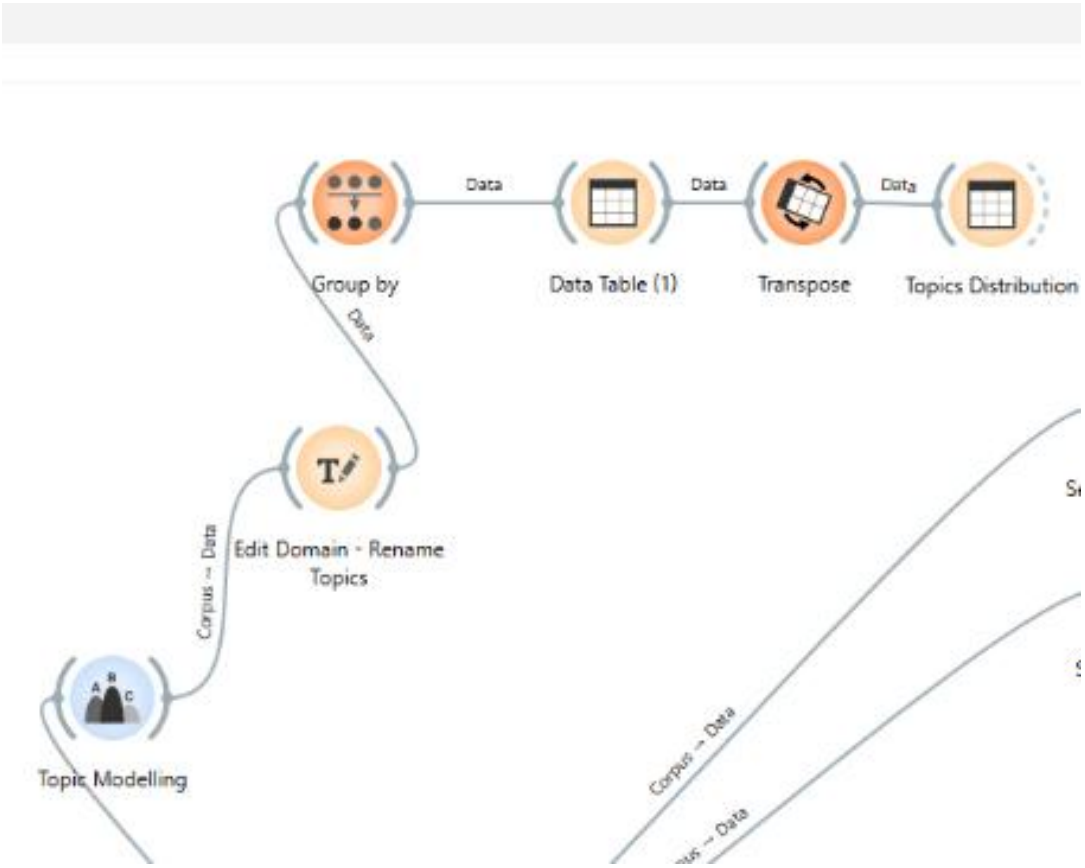
2.3 Branch B: Sentiment Analysis

Node	Purpose and Configuration
Sentiment Analysis	Applied VADER (Valence Aware Dictionary and sEntiment Reasoner) to each document. Produced four scores per article: positive (pos), negative (neg), neutral (neu), and compound (aggregate, range −1 to +1).
Group by + Transpose + Data Table	Grouped sentiment scores by class label (Fake/True) and transposed the result to produce a summary comparison table showing summed pos, neg, and neutral values per class.



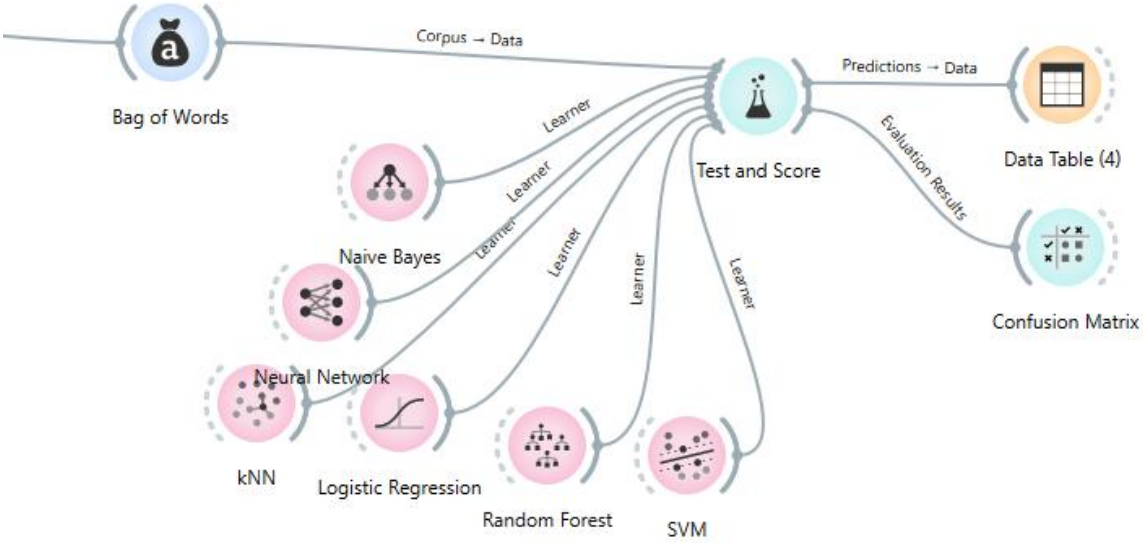
2.4 Branch C: Topic Modeling

Node	Purpose and Configuration
Topic Modeling	Applied Latent Dirichlet Allocation (LDA) with 5 topics, 100 iterations, and 10 keywords per topic. Output includes per-document topic probability scores.
Edit Domain - Rename Topics	Renamed generic topic labels (Topic 1-5) to descriptive names based on keyword inspection: Breaking Political News & Party Conflict; Election Integrity & Legislative Controversy; Social Media & Viral Political Content; Anti-Establishment & Identity Politics Narratives; Media Criticism & Cultural Commentary.
Group by + Transpose + Data Table	Aggregated topic probability sums by class label to compare topic dominance in Fake vs. True news.



2.5 Branch D: Supervised Classification

Node	Purpose and Configuration
Bag of Words	Generated TF-IDF document-term matrix (sublinear TF, smooth IDF) from the preprocessed corpus. Output: Embeddings channel.
Naive Bayes	Probabilistic baseline classifier. Default settings.
Neural Network	Default hidden layer configuration.
kNN	k-Nearest Neighbors. Default k-5. Included for comparative completeness.
Logistic Regression	L2 regularization, C=1.0
Random Forest	1000 trees, Replicable Training
Test and Score	10-fold stratified cross-validation. Metrics: Accuracy (CA), AUC, F1, Precision, Recall, MCC, Specificity, Log Loss.
Confusion Matrix	Per-class prediction error analysis for each classifier.



3. Algorithm Selection: Why These Classifiers?

Five classifiers were selected for this project, representing four distinct algorithmic families. The selection strategy was comparative: rather than committing to a single approach, I aimed to evaluate a range of model types - probabilistic, linear, ensemble, instance-based, and neural - to identify which best captures the patterns distinguishing fake from true political news. All classifiers received identical TF-IDF input features, enabling direct performance comparison.

Algorithm	Family	Rationale for Selection
Naive Bayes	Probabilistic	Standard baseline for text classification. Assumes feature independence, which holds reasonably well for bag-of-words representations. Computationally efficient and interpretable. Establishes the performance floor.
Logistic Regression	Linear / Discriminative	Well-suited to high-dimensional sparse TF-IDF vectors. L2 regularization prevents overfitting. Produces probability scores, supports AUC computation, and is interpretable via feature coefficients. Expected strong performer for text classification.
Random Forest	Ensemble / Tree-based	Handles high-dimensional feature spaces without feature scaling. Provides implicit feature importance rankings, revealing which vocabulary tokens are most predictive. Robust to irrelevant features through random subspace selection.
Neural Network	Neural / Non-linear	Included to test whether non-linear representations offer any gain over linear models on this text corpus. Capable of learning complex interactions between features, though may overfit on a corpus of this size without regularization tuning.
kNN (k-Nearest Neighbors)	Instance-based	Included primarily for comparative completeness. kNN classifies documents by proximity to their nearest labeled neighbors in TF-IDF space.

The inclusion of kNN was intentional: I anticipated it would underperform relative to the linear and ensemble models, and this expectation provides a useful contrast in the results discussion. Similarly, SVM with a linear kernel was part of the original plan but exhibited anomalous behavior (near-random accuracy of 53.6%) likely due to Orange's default RBF kernel setting rather than a linear kernel - a finding that is addressed in the lessons learned section.

4. Data: Source, Structure, and Preparation

4.1 Dataset Selection

Two publicly available labeled datasets were reviewed for this project. The primary dataset selected was the True and Fake News Sample, a balanced corpus of 44,867 political news articles drawn from the Kaggle fake news detection community. The secondary dataset reviewed was the LIAR Benchmark (Wang, 2017), a collection of approximately 12,800 short political statements from PolitiFact labeled across six veracity classes.

The True and Fake News Sample was selected over the LIAR Benchmark for three reasons: it contains full article body text (enabling richer sentiment and topic analysis), it uses a binary True/False label (simpler for classification evaluation), and its content directly covers the 2016-2018 U.S. election period that is the focus of this project.

4.2 Dataset Structure

Column	Description
title	Headline of the news article. Used as document identifier in Orange Corpus.
text	Full body text of the article. Primary text feature used in all analyses.
subject	Topic category from source
date	Publication date. Covers primarily 2016 - 2018.
label	Binary class variable added during preparation:

4.3 Data Preparation

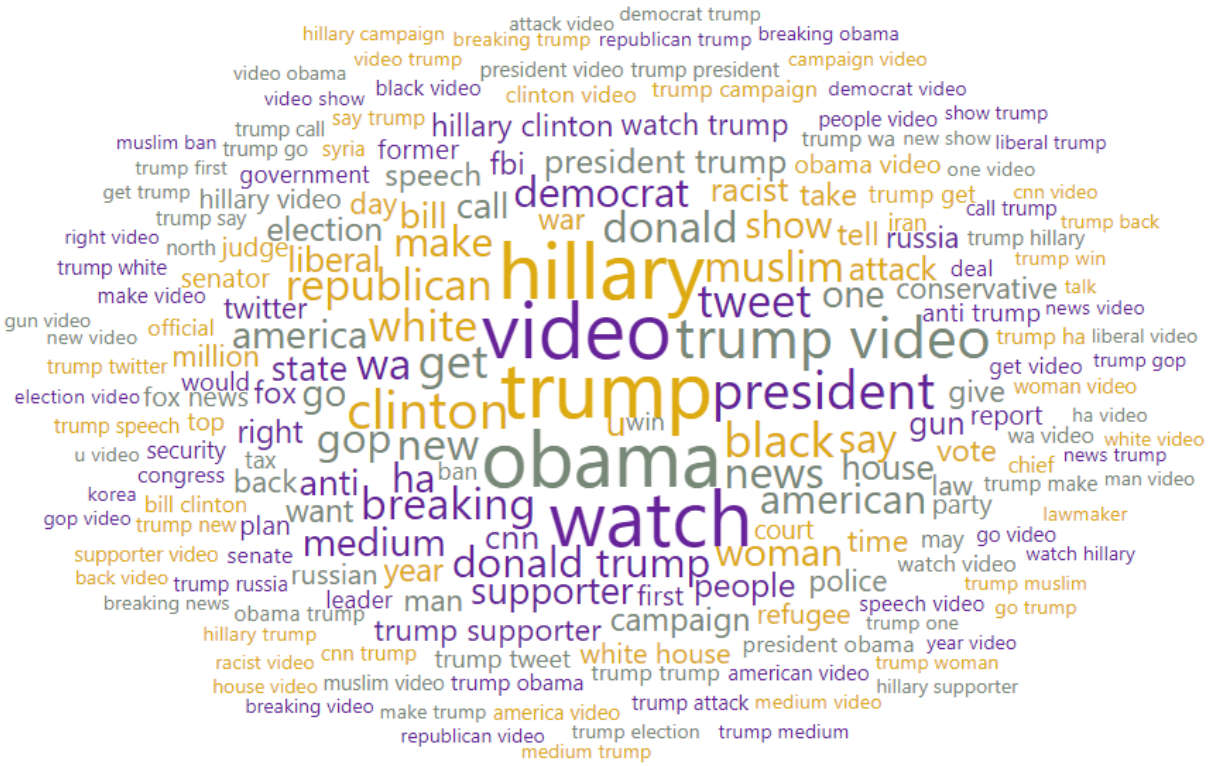
The two source files (True news sample.csv and Fake news sample.csv) were provided separately and required combination before import into Orange. The following steps were performed outside Orange:

- **Added a 'label' column** to each file: 'True' for all rows in the True file, 'Fake' for all rows in the Fake file.
- **Merged both files** into a single CSV (fake_news_combined.csv) by appending the Fake rows below the True rows, ensuring consistent column headers.
- **Verified balance:** True (21,417) and Fake (23,450) articles, total 44,867 rows with no missing values in the 'text' or 'label' columns.
- **No additional cleaning** was required outside Orange. All text normalization (lowercasing, tokenization, stopword removal, lemmatization) was handled within the Preprocess Text node.

5. Results: What the Data Revealed

The analysis was conducted against the backdrop of a historically significant media environment - the 2016 U.S. presidential election and its immediate aftermath. The findings across all four branches converge on a consistent picture: fake news from this period was not merely inaccurate - it was linguistically distinct, emotionally polarized, and topically concentrated in ways that made it identifiable through text mining alone.

5.1 Word Cloud Analysis: Vocabulary Fingerprints



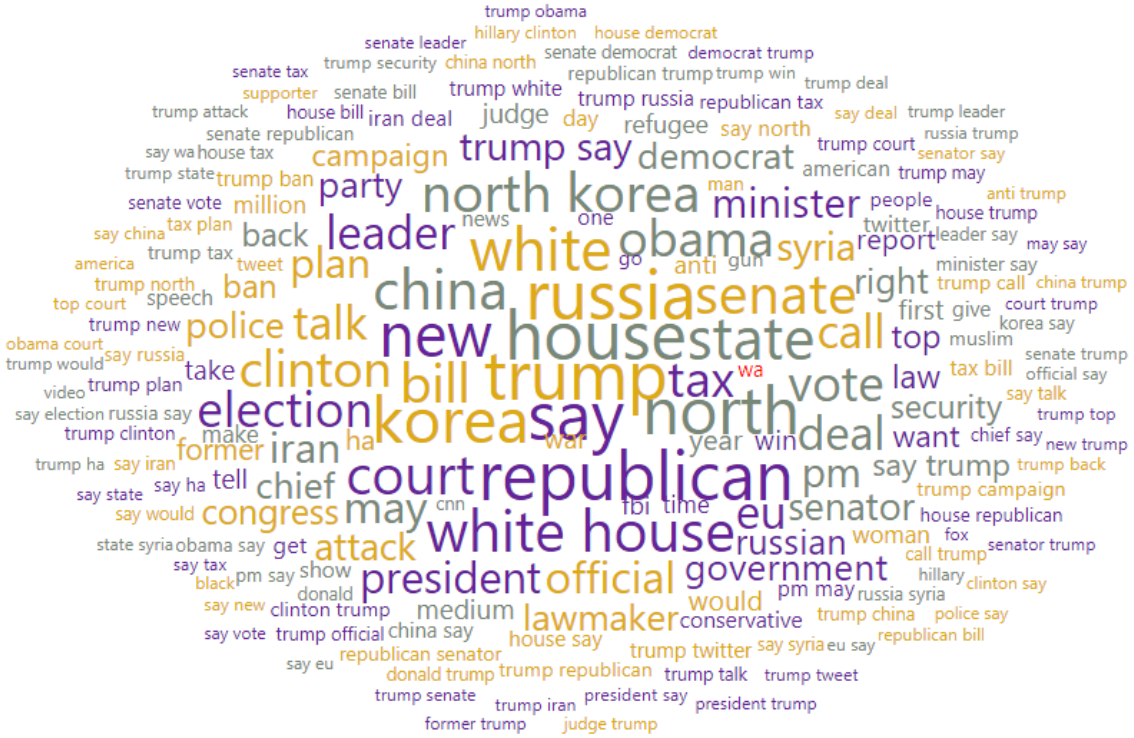


Figure 2. Word clouds for Fake news (above) and True news (below). Word size reflects token frequency.

The word clouds reveal very different vocabulary priorities between the two classes. In the fake news corpus, the dominant terms are hillary, video, trump, obama, watch, breaking, muslim, and president. This vocabulary pattern reflects several hallmarks of political misinformation: the heavy use of personal names (particularly political opponents) over institutional references, the prominence of action words like watch and breaking that create urgency and emotional arousal, and the inclusion of identity-coded terms like muslim that signal the identity-politics framing common in partisan misinformation.

The true news corpus, by contrast, is anchored by institutional and geopolitical vocabulary: house (as in White House), russia, korea, north, republican, election, senate, bill, and law. These are the terms of governance and international affairs as reported by institutional journalism. The word trump appears in both clouds - Trump was the dominant subject of political news in this period - but in true news it is surrounded by policy and institutional context, while in fake news it appears alongside emotionally charged terms.

The presence of video, watch, and tweet in the fake news cloud is particularly telling. These terms suggest that fake news articles from this period were frequently framed as media events - articles that referenced social media content, viral videos, or Twitter posts as their primary 'evidence.' This is consistent with the

known strategy of election-era misinformation sites that aggregated social media reactions rather than reporting primary-source facts.

5.2 Topic Modeling: Thematic Structure of the Corpus

Selected label	Feature name	Feature 1 No Fake	Feature 2 No True
1	Breaking Political News & Party Conflict - Sum	4036.24	4072.83
2	Election Integrity & Legislative Controversy - Sum	3123.55	3576.32
3	Social Media & Viral Political Content - Sum	6933.37	3593.59
4	Anti-Establishment & Identity Politics Narratives - Sum	5316.38	4887.18
5	Media Criticism & Cultural Commentary - Sum	4040.46	5287.08

Figure 3. Topic distribution by class label. Sums represent total topic probability mass allocated to each class.

LDA identified five topics across the combined corpus. The table below maps each topic to its keyword set, assigned label, and its distribution between Fake and True news.

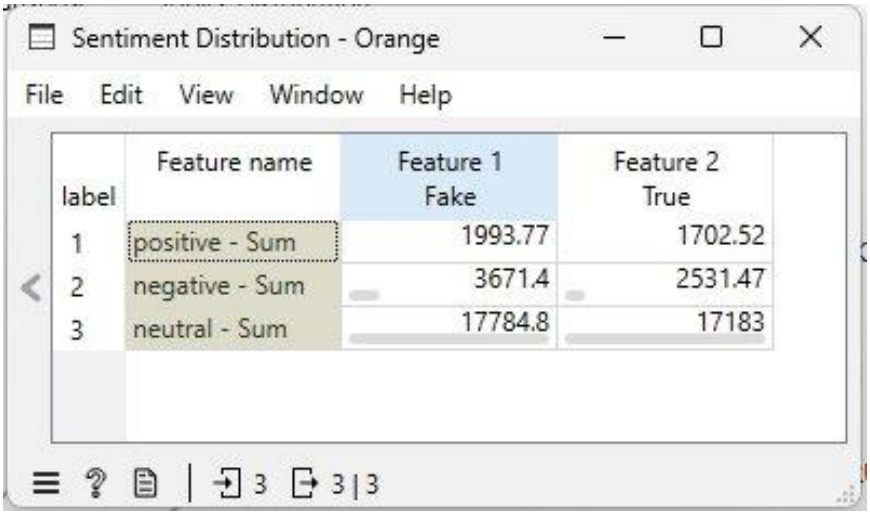
Topic Label	Top Keywords	Fake Sum	True Sum
Breaking Political News & Party Conflict	watch, breaking, trump, democrat, fbi, republican, gop	4,036	4,073
Election Integrity & Legislative Controversy	hillary, clinton, vote, bill, racist, law, security	3,124	3,576
Social Media & Viral Political Content	video, trump, white, liberal, donald, supporter	6,933	3,594
Anti-Establishment & Identity Politics	obama, trump, president, muslim, anti, police, america	5,316	4,887
Media Criticism & Cultural Commentary	cnn, medium, black, woman, war, say	4,040	5,287

The most revealing finding in the topic analysis is the distribution of Topic 3: Social Media and Viral Political Content. Fake news articles accumulated a total topic probability sum of 6,933 on this topic, compared to only 3,594 for true news - a ratio of nearly 2:1. This topic is characterized by vocabulary associated with

viral video content, supporter culture, and social media amplification. The finding directly maps to the known mechanics of 2016 election-era misinformation: fake news was disproportionately built around social media artifacts - tweets, Facebook videos, reaction posts - rather than primary news events.

Topic 5 (Media Criticism and Cultural Commentary) shows the opposite pattern: true news outscores fake news (5,287 vs. 4,040). This reflects the dominance of institutional journalism - Reuters, AP, major outlets - in the true news corpus, which frequently covered issues of media literacy, racial equity, and foreign affairs from a factual, analytical perspective. Topics 1 and 4 (Breaking Political News and Anti-Establishment Narratives) are relatively balanced between classes, indicating that both fake and true news covered similar political subjects - but, as the word cloud analysis shows, they covered them using fundamentally different language.

5.3 Sentiment Analysis: Emotional Tone by Class



label	Feature name	Feature 1 Fake	Feature 2 True
1	positive - Sum	1993.77	1702.52
2	negative - Sum	3671.4	2531.47
3	neutral - Sum	17784.8	17183

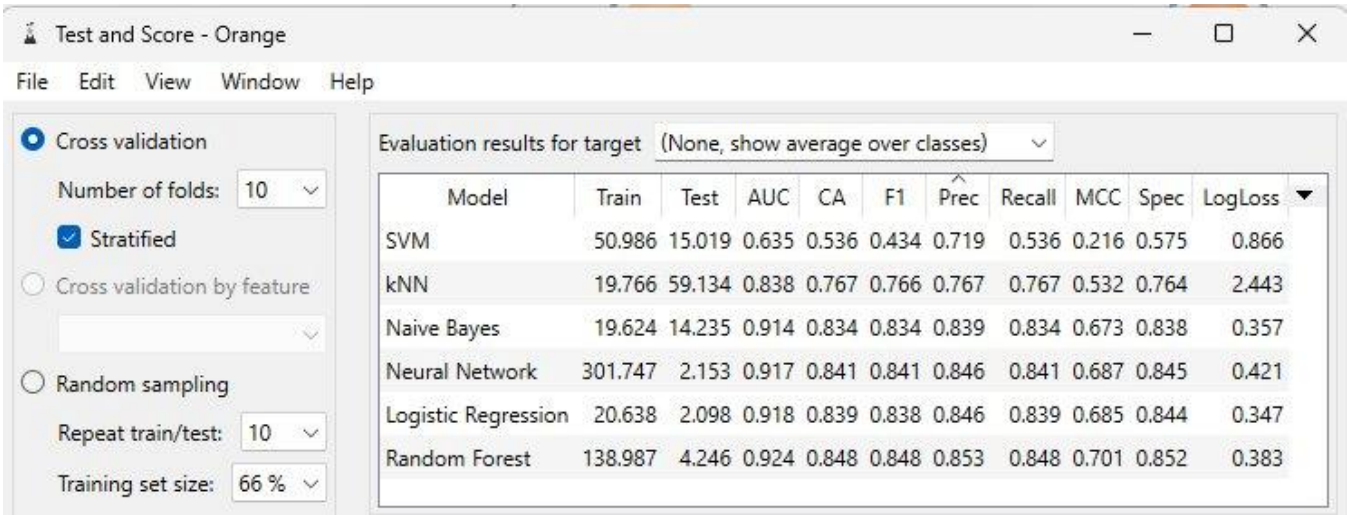
Figure 4. VADER sentiment scores summed by class label. Values represent aggregated positive, negative, and neutral scores across all articles per class.

The sentiment distribution table confirms a pattern consistent with research on political misinformation: fake news articles carry systematically higher emotional charge in both the positive and negative directions. Fake news articles scored a total negative sentiment sum of 3,671 compared to 2,531 for true news - an increase of approximately 45%. On the positive dimension, fake news also outscored true news (1,994 vs. 1,703), suggesting not merely a negativity bias but a broader emotional amplification strategy.

Neutral sentiment, which reflects factual, measured, or balanced language, was slightly higher in fake news (17,785) than true news (17,183). This is not contradictory: on a document-level basis, VADER averages sentiment across all sentences in an article, including factual transitional sentences. The presence of high neutral scores in both classes simply reflects the length of the articles. What distinguishes the classes is the relative proportion of charged language: fake news shows a higher ratio of positive + negative to neutral than true news, indicating that fake articles contain more emotionally loaded sentences interspersed with otherwise neutral prose.

The election-era context is critical for interpreting these numbers. The 2016 campaign was characterized by unusually high emotional polarization - both online and in media coverage. Fake news amplified this polarization deliberately, deploying outrage, fear, and partisan triumphalism as engagement drivers.

5.4 Predictive Classification: Algorithm Performance



Model	Train	Test	AUC	CA	F1	Prec	Recall	MCC	Spec	LogLoss
SVM	50.986	15.019	0.635	0.536	0.434	0.719	0.536	0.216	0.575	0.866
kNN	19.766	59.134	0.838	0.767	0.766	0.767	0.767	0.532	0.764	2.443
Naive Bayes	19.624	14.235	0.914	0.834	0.834	0.839	0.834	0.673	0.838	0.357
Neural Network	301.747	2.153	0.917	0.841	0.841	0.846	0.841	0.687	0.845	0.421
Logistic Regression	20.638	2.098	0.918	0.839	0.838	0.846	0.839	0.685	0.844	0.347
Random Forest	138.987	4.246	0.924	0.848	0.848	0.853	0.848	0.701	0.852	0.383

Figure 5. Test and Score results - 10-fold stratified cross-validation across five classifiers.

The classification results show strong performance from four of the five algorithms, with one notable outlier. The table below summarizes the key metrics extracted from the Test and Score widget:

Model	AUC	Accuracy	F1	Precision	Recall	MCC
Random Forest	**0.924**	**84.8%**	**0.848**	**0.853**	**0.848**	**0.701**
Logistic Regression	0.918	83.9%	0.838	0.846	0.839	0.685
Neural Network	0.917	84.1%	0.841	0.846	0.841	0.687
Naive Bayes	0.914	83.4%	0.834	0.839	0.834	0.673
kNN	0.838	76.7%	0.766	0.767	0.767	0.532
SVM	0.635	53.6%	0.434	0.719	0.536	0.216

Random Forest achieved the highest overall performance, with an AUC of 0.924 and classification accuracy of 84.8%. This means the model correctly identified approximately 848 out of every 1,000 articles as Fake or True. The AUC of 0.924 indicates strong discriminative ability - the model can reliably rank fake articles above true articles across a wide range of classification thresholds, which is precisely the capability needed for real-world content moderation where different confidence thresholds might be applied for different use cases (e.g., auto-flagging vs. human review queues).

Logistic Regression and Neural Network performed nearly identically (AUC 0.918 and 0.917 respectively), reinforcing the finding from the text classification literature that linear models trained on TF-IDF representations are highly competitive with non-linear approaches on this type of problem. The similarity in performance between Logistic Regression and Neural Network also suggests that the distinguishing vocabulary patterns in this dataset are largely linearly separable - the TF-IDF space has sufficient structure that non-linear boundaries do not provide additional lift. Naive Bayes, despite its simplicity, performed within two percentage points of the top models (AUC 0.914), confirming its well-known effectiveness in text classification tasks.

kNN showed a meaningful performance drop (AUC 0.838, accuracy 76.7%), consistent with its known weakness in high-dimensional sparse feature spaces. In a TF-IDF matrix with potentially thousands of features, distance measures become unreliable - a phenomenon known as the curse of dimensionality - and the most informative dimensions are drowned out by the many near-zero dimensions shared across all documents.

SVM produced near-random results (AUC 0.635, accuracy 53.6%). This is almost certainly attributable to a configuration error rather than a limitation of the algorithm: Orange 3.39's SVM widget defaults to an RBF (radial basis function) kernel, which is computationally prohibitive and performs poorly on sparse TF-IDF data. A linear kernel - the standard for text classification - would likely have produced results comparable to or exceeding Logistic Regression. This is documented in the lessons learned section as a key finding.

6. Lessons Learned: The Three W's

6.1 What Went Well

- **The workflow architecture was clean and reproducible.** Orange's node-based canvas made it straightforward to build four parallel analytical branches from a single preprocessed corpus. Every configuration decision is visible in the workflow screenshot, making the project fully replicable. Unlike other tools like KNIME (was limited in node configuration) and Rapidminer (had a limitation on rows)
- **Topic modeling produced strong, interpretable results.** The five LDA topics aligned well with known themes from the 2016 election cycle. The renaming via the Edit Domain widget made outputs meaningful without requiring code.
- **The sentiment and topic findings told a consistent story.** The combination of higher emotional charge in fake news (sentiment) and disproportionate social-media-oriented topic loading (Topic 3) gave a true interpretation of how fake news was constructed and distributed during the election period.
- **Four of five classifiers achieved strong accuracy (AUC > 0.91).** The top four models all exceeded the 0.90 AUC threshold, providing strong evidence that the vocabulary and topical patterns in this dataset are sufficient to support automated fake news detection with meaningful real-world reliability.

6.2 What Did NOT Go Well

- SVM failed due to an incorrect kernel setting. The default RBF kernel in Orange's SVM widget is inappropriate for sparse TF-IDF data. The near-random accuracy (53.6%) misrepresents the true capability of SVM on this type of problem. The fix - switching to a linear kernel - was identified at a later stage in the project.
- **The word 'ha' and noise tokens survived preprocessing.** Despite stopword filtering and minimum length constraints, some non-content tokens (such as 'ha', 'wa', 'u') appear in both word clouds. These are likely artifacts of OCR or informal text encoding in the source data and should have been addressed with a custom stopword list.
- **The dataset is temporally narrow.** Most articles are from 2017 through 2018. While this makes the dataset coherent for election-cycle analysis, it means the trained models may not generalize well to fake news from different time periods, topics, or geopolitical contexts.
- **The N-**gram node in the topic modelling and sentiment analysis sections produced a lot of noise and didn’t produce any meaningful results, as such I had to exclude it because it needed a lot of fine tuning and I was not able to do it given the time frame for the project.

6.3 What Would Be Done Differently Next Time

- Fix the SVM kernel before running classification. Explicitly set the SVM kernel to Linear in Orange before connecting it to Test and Score. This is a single setting change that would likely push SVM accuracy and provide a more complete comparison across model families.
- **Build a custom stopword list.** Create a project-specific stopword file that adds informal tokens, single-character artifacts, and URL fragments.
- **Expand the dataset across time periods.** Include articles from other election cycles to test whether models trained on 2016-era fake news patterns generalize to subsequent election cycles or require retraining - an important practical question for ongoing content moderation.
- **Explore bigrams and trigrams in Bag of Words.** Adding an N-grams Range preprocessor node would allow the classifier to learn from phrases like 'fake news', 'deep state', or 'rigged election' rather than individual tokens - capturing the specific word combinations that characterized election-era misinformation.